Dry Run (cohort 1) x

[2, 2, 1, 2, 2]

(K = 3) (ans = 2)

[2, 0, 0, 1, 2, 0]

tar = 10

[2, 5, 10, 1, 30]

→ Linear Search

→ Maximum in Array

→ Min^m in Array

→ Reverse Array

→ Reverse Array Range

→ 3 dry Run

→ (cohort 1)

arr(i) - , , tar, [10]

i = 1 arr(i)

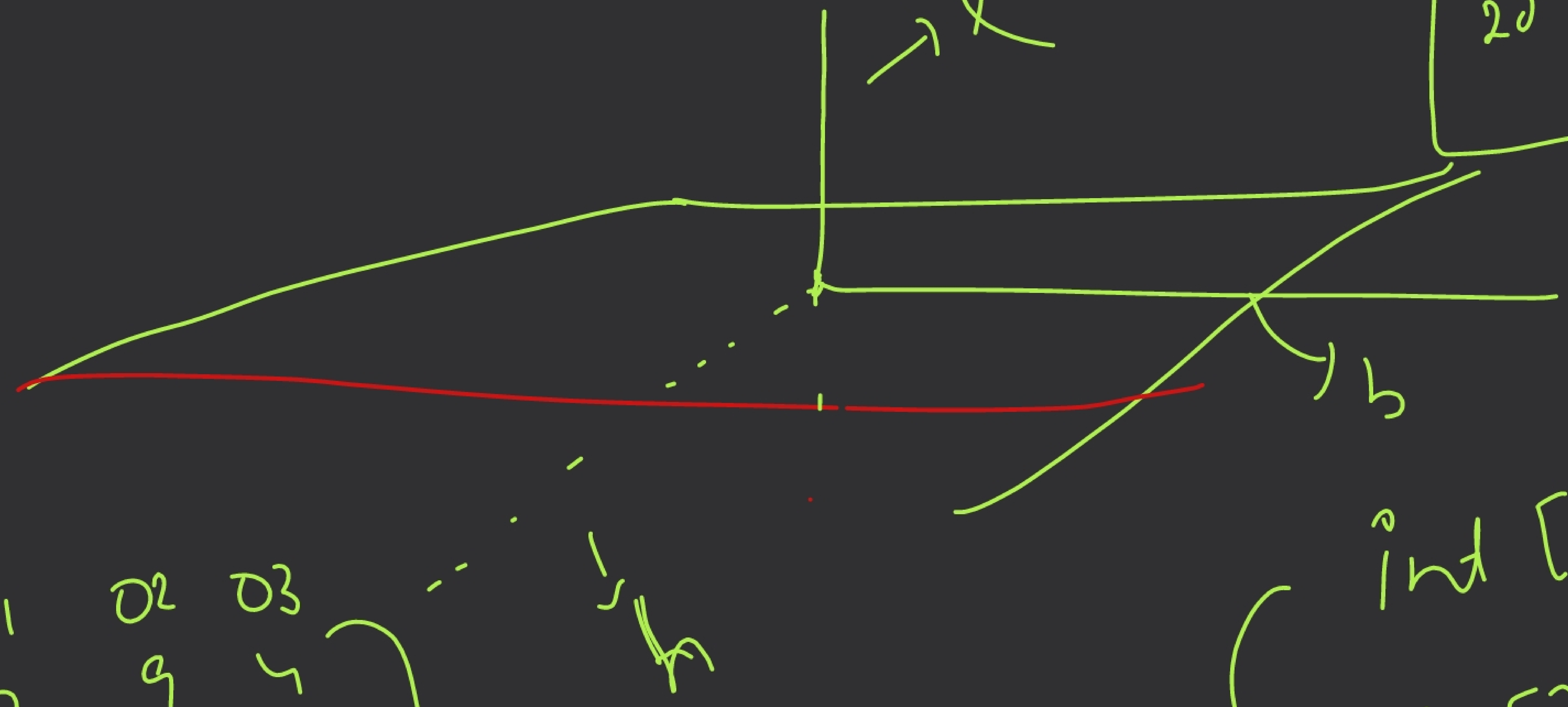
i = 1

2D array :-

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array}$$

1st $\begin{bmatrix} 2 & 3 & 4 & 5 & 6 \end{bmatrix}$

$$\begin{bmatrix} (00) & (01) & (02) \\ 5 & 7 & 9 \\ (10) & (11) & (12) \\ 10 & 12 & 13 \\ (20) & (21) & (22) \\ 2 & 8 & 10 \end{bmatrix}$$



$$\begin{array}{cccc} 00 & 01 & 02 & 03 \end{array}$$
$$\begin{bmatrix} 5 & 7 & 9 & 4 \\ & 11 & 12 & 13 \\ 10 & 8 & 6 & 8 \\ 2 & & & \end{bmatrix}$$

$$\begin{array}{c|cccc} & 00 & \cdots & & 03 \\ \hline & x & y & z & 100 \\ & a & b & c & 8 \\ 20 & A & e & f & 3 \end{array}$$

25

```
int [] arr = new int[5];
int [][] arr = new int[3][4];
```

$$
\begin{array}{cccc}
00 & 01 & 02 & 03 \\
5 & 8 & 7 & 6 \\
10 & 11 & 12 & 13 \\
2 & 8 & 7 & 5 \\
20 & 21 & 22 & 23 \\
9 & 10 & 100 & 120
\end{array}
$$

n → Tot Rows

m → (col)

arr[i][j]

↓

(row) 4 (col)

↓

0 → 0, 1, 2, 3

1 → 0, 1, 2, 3

2 → 0, 1, 2, 3

1 2 3 4

00 01 02 03 10 11 23

(5 8 7 6 10 11 . . . . 120)

(5 10 2 20 . . . . . 120)

Output

```
for(i=0; i<n; i++){
    for(j=0; j<m; j++){
        arr[i][j] = sc.nextInt();
    }
}
```

i = 0

j = 0 1 2 3

//Output

```
for( i=0; i<n; i++){
    for( j=0; j<m; j++){
        System.out.print( arr[i][j] +
                " ");
    }
    System.out.println();
}
```

Row wise Traversal :

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

[1 2 3 4]

Column Wise Traversal : [1 3 2 4]

00 01 10 11

00 10 01 11

1 3 2 4

arr(j)(i)

```
for(i=0; i<n; i++){
    for(j=0; j<m; j++){
        S.o.p(arr(i)(j)+" ");
    }
}
```

Wave Print

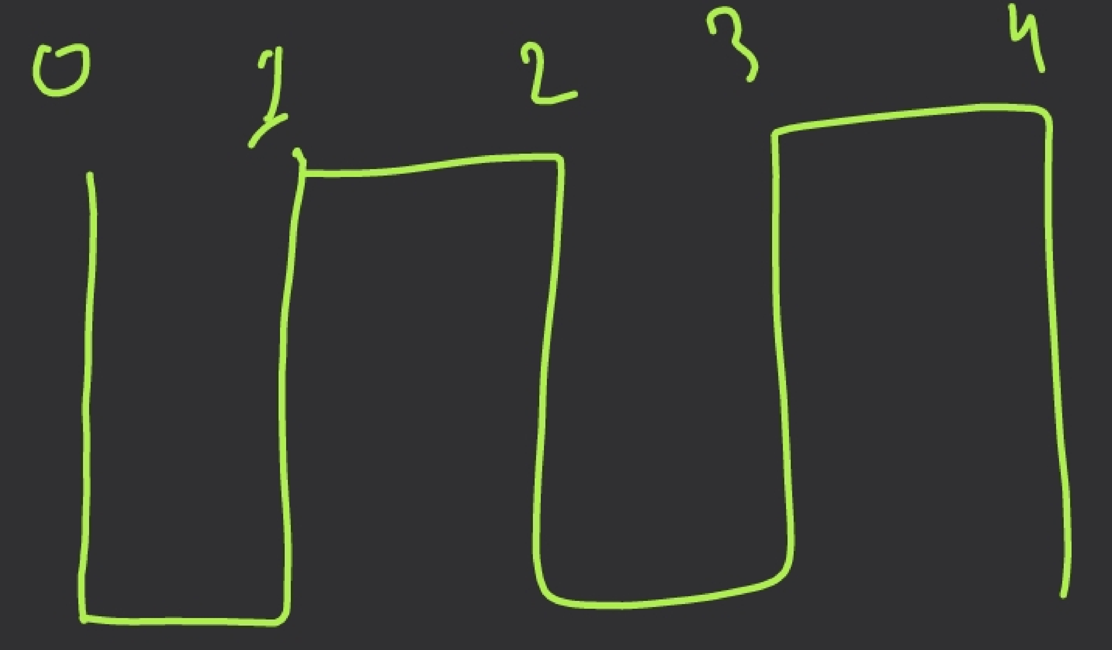
$$\begin{array}{ccc} 00 & 01 & 02 \\ 1 & 2 & 3 \\ 10 & 11 & 12 \\ 4 & 5 & 6 \\ 20 & 21 & 22 \\ 7 & 8 & 9 \end{array}$$

1 4 7 8 5 2 3 6 9

00 10 20 21 11 01 02 12 22

Col = 0 Col = 1 Col = 2

Steps to solve 2D array problems:-



→ Write all the index numbers

→ Analyze the constant & variable indexes

→ Make some relation among variable (even → T to B, odd → B to T)

→ Do write the code. (before writing write the possible output on notebook).

$$\begin{bmatrix} \overset{00}{1} & \overset{01}{2} & \overset{02}{3} \\ \overset{10}{4} & \overset{11}{5} & \overset{12}{6} \\ \overset{20}{7} & \overset{21}{8} & \overset{22}{9} \end{bmatrix}$$

(H.W.

)

```
for(j=0; j<m; j++) {
if(j%2==0) {
for(i=0; i<n; i++){
    S.O.P.(arr[i][j]+" ");
}
else {
    for(i=n-1; i>=0; i--){
        S.O.P.(arr[i][j]+" ");
    }
}
```

Transpose :-

$$\begin{array}{ccc} 00 & 01 & 02 \\ \end{array}$$
$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{\text{Transpose}} \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$(n \times n)$

```
for(i=0→m){
 for(j=0→m){
  transp[i][j] = arr[j][i];
 }
}
```

$$\rightarrow \begin{array}{ccc} 00 & 10 & 20 \\ 01 & 11 & 21 \\ 02 & 12 & 22 \end{array}$$

$(n \neq m)$

trans[0][0] = arr[0][0]
tran[0][1] = arr[1][0]
tran[0][2] = arr[2][0]
tran[1][0] = arr[0][1]
tran[1][1] = arr[1][1]
tran[1][2] = arr[2][1]
tran[2][0] = arr[0][2]
tr[2][1] = arr[1][2]
tran[2][2] = arr[2][2]

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \Longrightarrow \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

→ row < m

→ row < n

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

```
for(int i=0; i<n; i++){
    for( j=i+1; j<n; j++){
        swap(arr[i][j], arr[j][i]);
    }
}
```